

Macroeconomics III (4,212)

Exam Compilation

2019

1. Describe the main empirical evidence supporting and contradicting the basic Solow model.
2. Explain the Solow model with **land and natural resources**. Explain how you would numerically simulate the transition equations with a software of your choice (Excel, or other).
3. Derive and discuss the policy implications of Cozzi (2017) hybrid model.
4. **What are the causes of structural unemployment?**

2020

1. Describe the main empirical evidence supporting and contradicting the basic Solow model.
2. Explain the Solow model with **land and natural resources**. Sketch how you would numerically simulate the transition equations with a software of your choice (Excel, or other).
3. Derive and discuss the policy implications of Cozzi (2017) hybrid model.
4. **Derive the “Solow conditions” of the efficiency wages model and give an intuitive explanation of them.**

2023

1. Describe qualitatively (no need to provide exact numbers) empirical evidence and stylized facts in favour of and against the semi-endogenous growth model with productive externalities.
2. Consider an economy already at the positive balanced growth path in the Solow growth model with **oil and land**. Suppose that the investment rate, s , suddenly and permanently increases (constant everything else): how would you simulate the consequences using a programme of your choice (Excel, etc)?
3. Explain Mankiw, Romer, and Weil’s growth model with human capital. Sketch how you would numerically simulate the transition equations with a software of your choice (Excel, or other).
4. **Based on the efficiency wage theory, how would the unemployment rate respond to an increase in unemployment benefits? And how would it respond to more competition among firms?**

Unknown 1

1. Carefully explain the most important empirical evidence in favour of the general Solow model with exogenous technological progress.
2. Describe analytically and graphically the transition equation for per-capita capital in the Solow model with human capital, and explain in detail how you would numerically simulate it with a software of your choice (Excel, or other).
3. Explain the economic difference between "endogenous growth" and "semi-endogenous growth".
4. State the main equations of the complete R&D "Romer" macromodel of semi-endogenous growth. Provide a careful economic explanation of these equations.

Unknown 2

1. Does the general Solow model with exogenous technological progress imply convergence to a balanced growth path? Why so or why not? Do empirical data support this prediction?
2. Describe analytically and graphically the transition equation for per-capita capital in the basic Solow model, and explain in detail how you would numerically simulate it with a software of your choice (Excel, or other).
3. Consider an economy already at the positive balanced growth path in the Solow growth model with **land and natural resources**. Suppose that the oil extraction rate, S_e , suddenly and permanently decreases (constant everything else): what happens to the level of per capita GDP? What will happen to the long-run growth rate of per capita GDP?
4. State the main equations of the complete R&D "Romer" macromodel of endogenous growth. Provide an economic explanation of these equations.